

AI Invoice Processing Agent

Project Documentation & Evidence Report

1. Project Overview

This project implements an Agentic AI-powered invoice processing workflow using n8n. It receives invoice PDFs through Gmail, stores the original invoice in Google Drive, extracts invoice text, uses Google Gemini to convert unstructured invoice information into structured JSON, validates the data, checks duplicates, retrieves purchase-order information, performs deterministic PO/vendor/amount/currency checks, makes a processing decision, logs the result in Google Sheets, and sends an appropriate Gmail notification.

Invoices requiring review are routed to a human instead of being automatically approved.

2. Business Problem

Manual invoice processing involves email intake, document reading, data entry, validation, purchase-order matching, duplicate checking, decisions, and record keeping. It can lead to data-entry errors, delays, duplicate processing, inconsistent checks, and limited traceability.

The project combines LLM-based document understanding with deterministic workflow automation and human review for exceptions.

3. Project Objective

Build an Agentic AI-powered invoice processing workflow that automatically receives invoice PDFs, extracts structured invoice information using an LLM, validates the data, detects duplicate invoices, matches invoices with purchase orders, applies deterministic business rules, records processing results, and routes exceptions to human review.

4. Final Workflow Architecture

Invoice Email Trigger

- ↓ Retrieve Invoice Email
- ↓ Extract Invoice Text
- ↓ AI Invoice Data Extraction
- ↓ Invoice Data Validation
- ↓ Check Duplicate Invoice
- ↓ Duplicate Invoice Detection
- ↓ Retrieve PO Details
- ↓ PO & Vendor Validation
- ↓ Invoice Decision Engine
- ↓ Log Invoice Result

↓ Route Invoice Decision

↙ Send Human Review Alert ↘ Send Processing Notification

Supporting components: the original PDF is stored in Google Drive from the email-retrieval stage. Google Gemini and the Structured Output Parser are connected to the AI extraction stage.

SCREENSHOT 1 — Final n8n Workflow

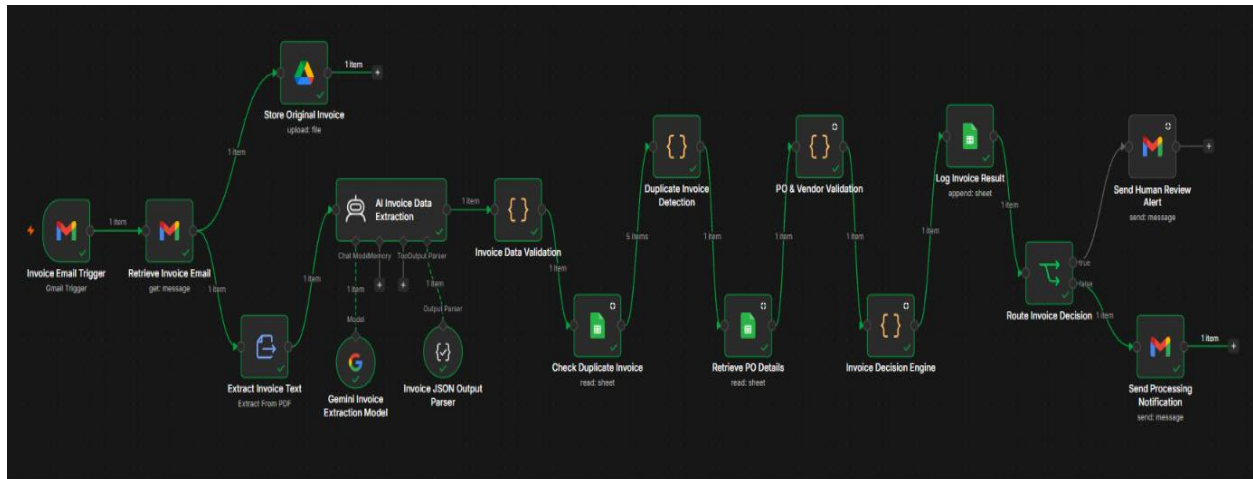


Figure 1. Final n8n Workflow

5. Technology Stack

n8n Cloud — workflow orchestration

Gmail — invoice trigger and notifications

Google Drive — original invoice storage

n8n Extract from File — PDF extraction

Google Gemini — invoice information extraction

Structured Output Parser — structured JSON

JavaScript / n8n Code — deterministic rules

Google Sheets — PO master and invoice log

Human-in-the-Loop — exception review

6. AI and Automation Design

The LLM is responsible for understanding unstructured invoice text and extracting required fields. Deterministic n8n rules handle validation, duplicate detection, PO matching, and the final decision. This keeps financial decision logic outside the LLM.

Extracted fields: invoice number, vendor name, invoice date, due date, subtotal, tax, total amount, currency, and PO number.

7. Data Sources

7.1 PO Master

PO-501 — ABC Supplies Pvt Ltd — ₹45,000 — INR — Approved

PO-502 — XYZ Traders Pvt Ltd — ₹30,000 — INR — Approved

PO-503 — Global Office Supplies — ₹75,000 — INR — Approved

PO-504 — Tech Solutions Pvt Ltd — ₹120,000 — INR — Approved

PO-505 — Office Mart India — ₹55,000 — INR — Approved

7.2 Invoice Log

Invoice_Log stores invoice number, vendor, dates, PO number, amounts, validation status/issues, PO match status, decision, reason, and processed date.

SCREENSHOT 2 — PO Master / Invoice Log

AI Invoice Processing Database					
File Edit View Insert Format Data Tools Gemini Extensions Help					
Q Menus ↶ ↷ 🖨 🔍 100% £ % .0 .00 123 Arial 10					
	A	B	C	D	E
1	PO_Number	Vendor_Name	PO_Amount	Currency	Status
2	PO-501	ABC Supplies Pvt Ltd	45000	INR	Approved
3	PO-502	XYZ Traders Pvt Ltd	30000	INR	Approved
4	PO-503	Global Office Supplies	75000	INR	Approved
5	PO-504	Tech Solutions Pvt Ltd	120000	INR	Approved
6	PO-505	Office Mart India	55000	INR	Approved
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+ ≡ PO_Master Invoice Log					

Figure 2. PO Master / Invoice Log

8. AI Invoice Data Extraction

The PDF is converted to text and passed to Gemini. The agent is instructed to extract only information present in the invoice, avoid guessing, normalize the invoice date to YYYY-MM-DD, identify INR, and return numeric amounts without currency symbols or commas. The Structured Output Parser enforces the JSON structure.

SCREENSHOT 3 — Gemini Invoice Extraction

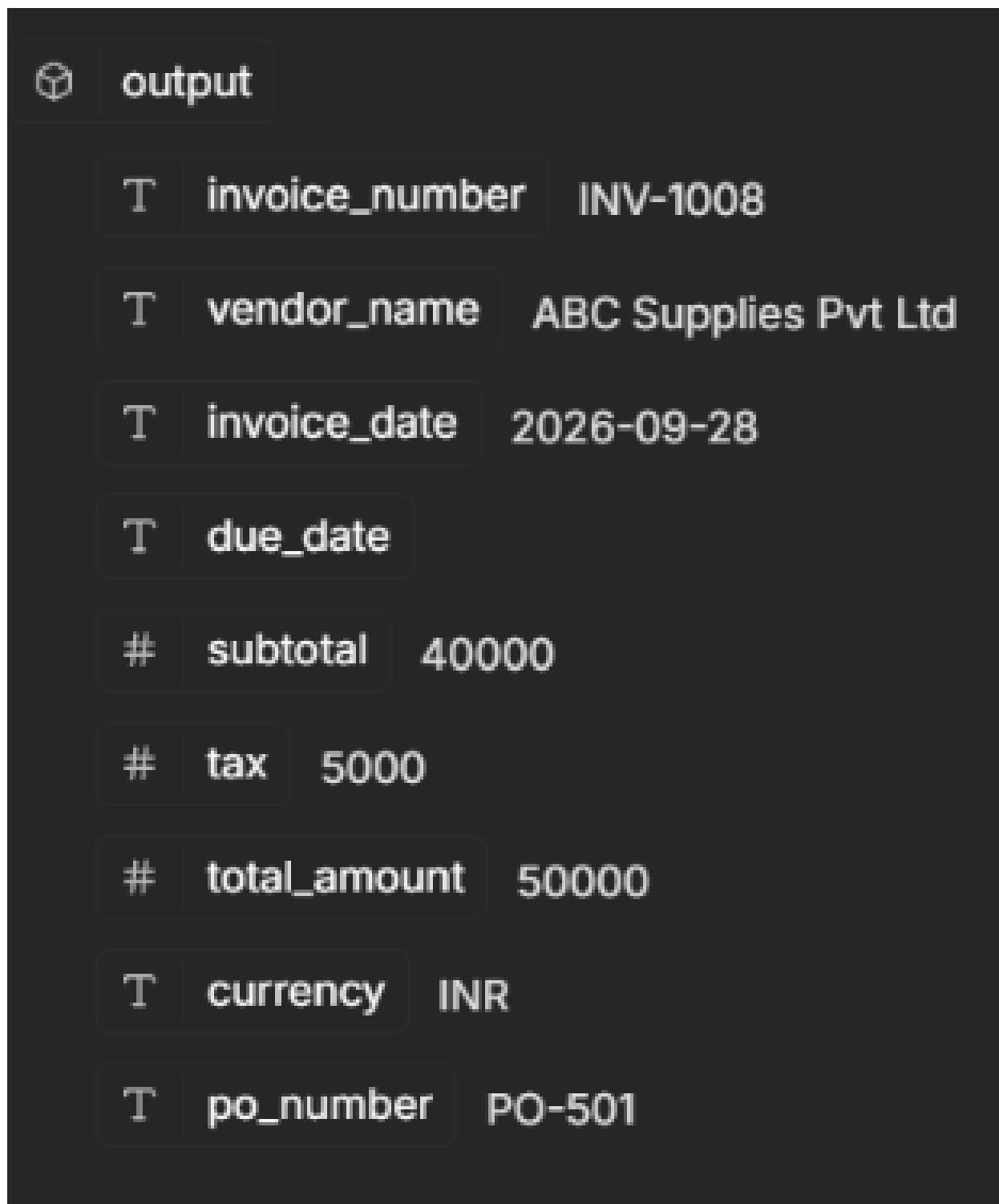


Figure 3. Gemini Invoice Extraction

9. Invoice Validation

Validation checks required fields and verifies $\text{subtotal} + \text{tax} = \text{total amount}$. Missing required values or arithmetic inconsistencies are recorded as validation issues and produce INVALID status.

SCREENSHOT 4 — Invoice Data Validation

OUTPUT		Schema	Table	JSON
1 item				
T	invoice_number	INV-1008		
T	vendor_name	ABC Supplies Pvt Ltd		
T	invoice_date	2026-09-28		
T	due_date			
T	po_number	PO-501		
#	subtotal	40000		
#	tax	5000		
#	total_amount	50000		
T	currency	INR		
T	validation_status	INVALID		
T	validation_issues	Amount mismatch: subtotal + tax = 45000, but invoice total = 50000		

Figure 4. Invoice Data Validation

10. Duplicate Invoice Detection

The workflow searches Invoice_Log by invoice number. If the number already exists, the invoice is marked DUPLICATE and the Decision Engine rejects it.

SCREENSHOT 5 — Duplicate Detection

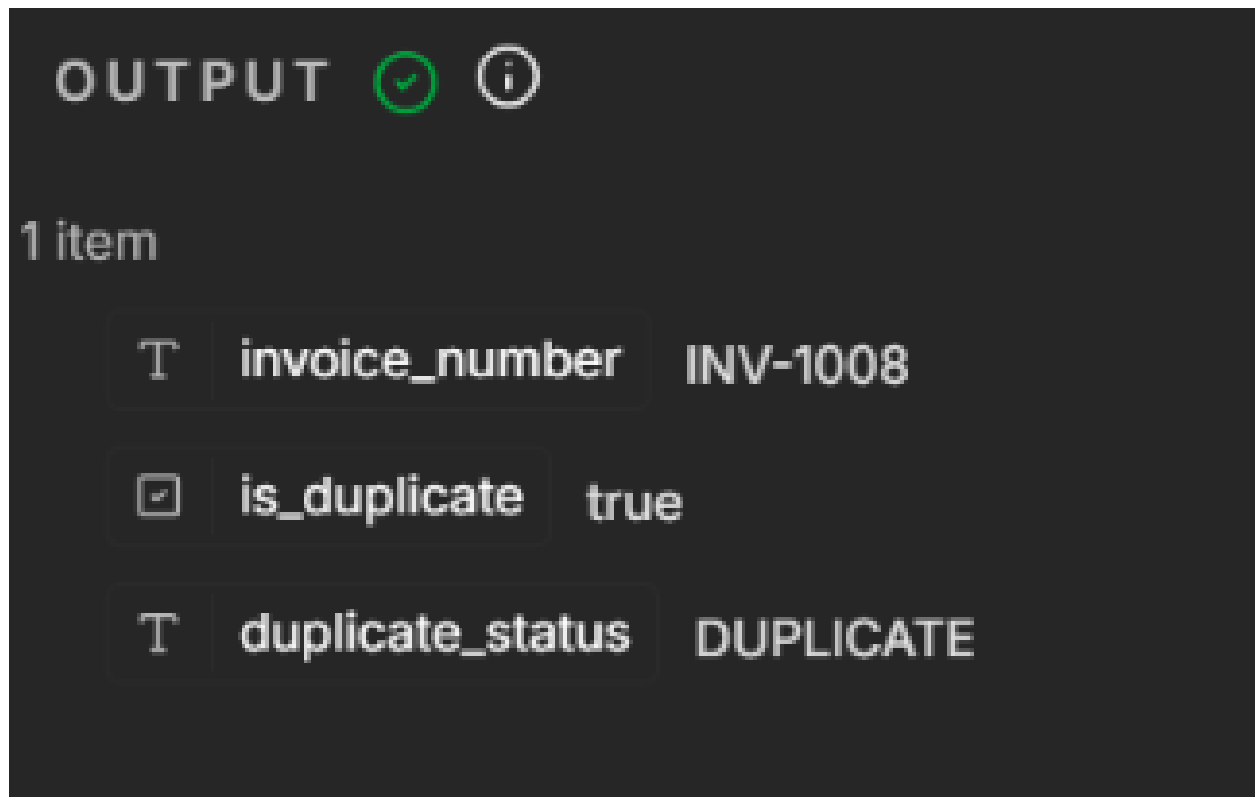


Figure 5. Duplicate Detection

11. PO and Vendor Validation

The workflow retrieves the PO using the invoice PO number and compares vendor, total amount, currency, and PO status. A mismatch is routed to human review.

SCREENSHOT 6 — PO & Vendor Validation

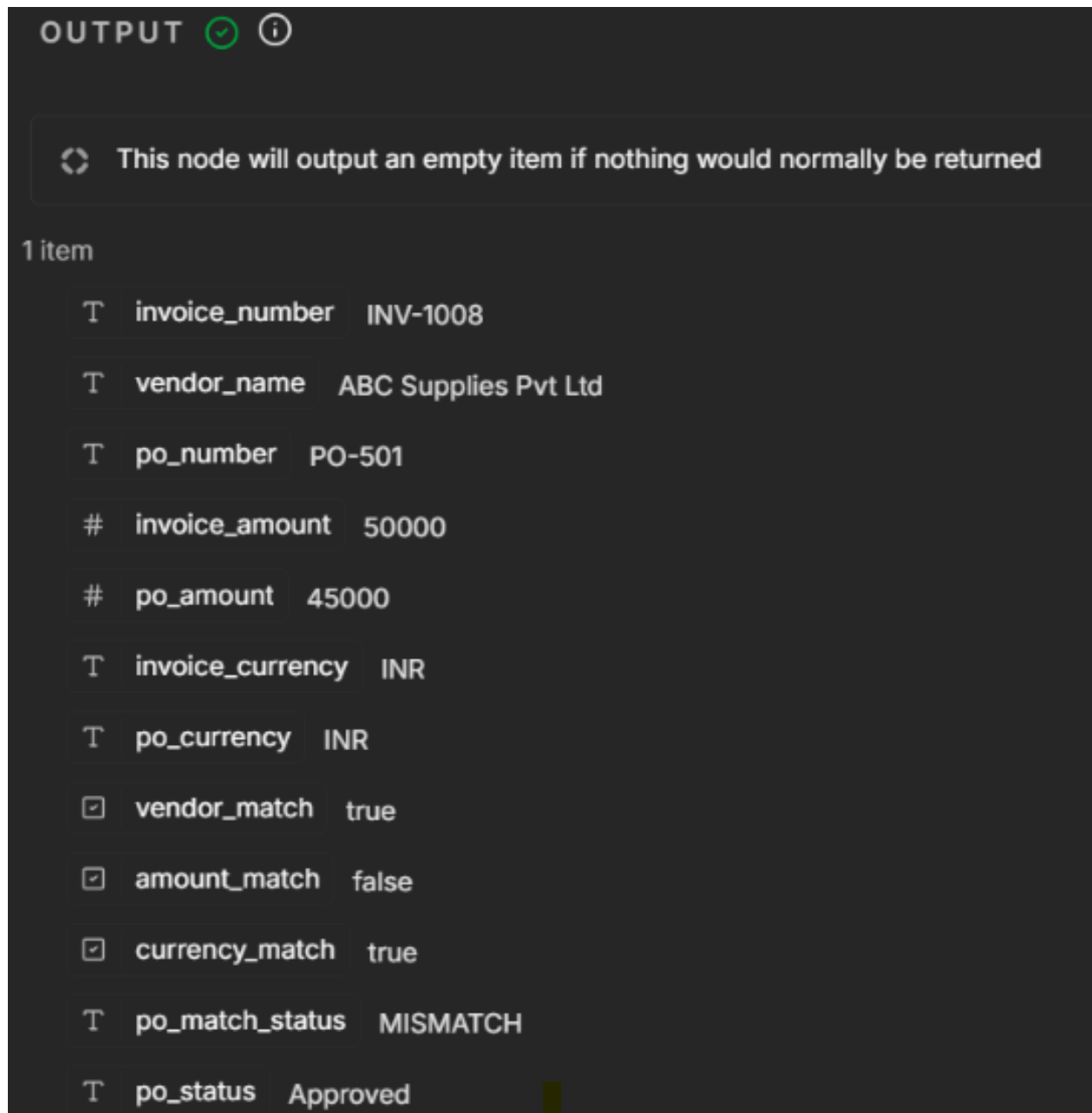


Figure 6. PO & Vendor Validation

12. Invoice Decision Engine

Duplicate invoice → REJECTED

Invoice validation failed → HUMAN REVIEW

PO/vendor/amount/currency matching failed → HUMAN REVIEW

PO is not Approved → HUMAN REVIEW

All checks pass → APPROVED

The decision reason is stored for traceability.

SCREENSHOT 7 — Decision Engine — Approved Case

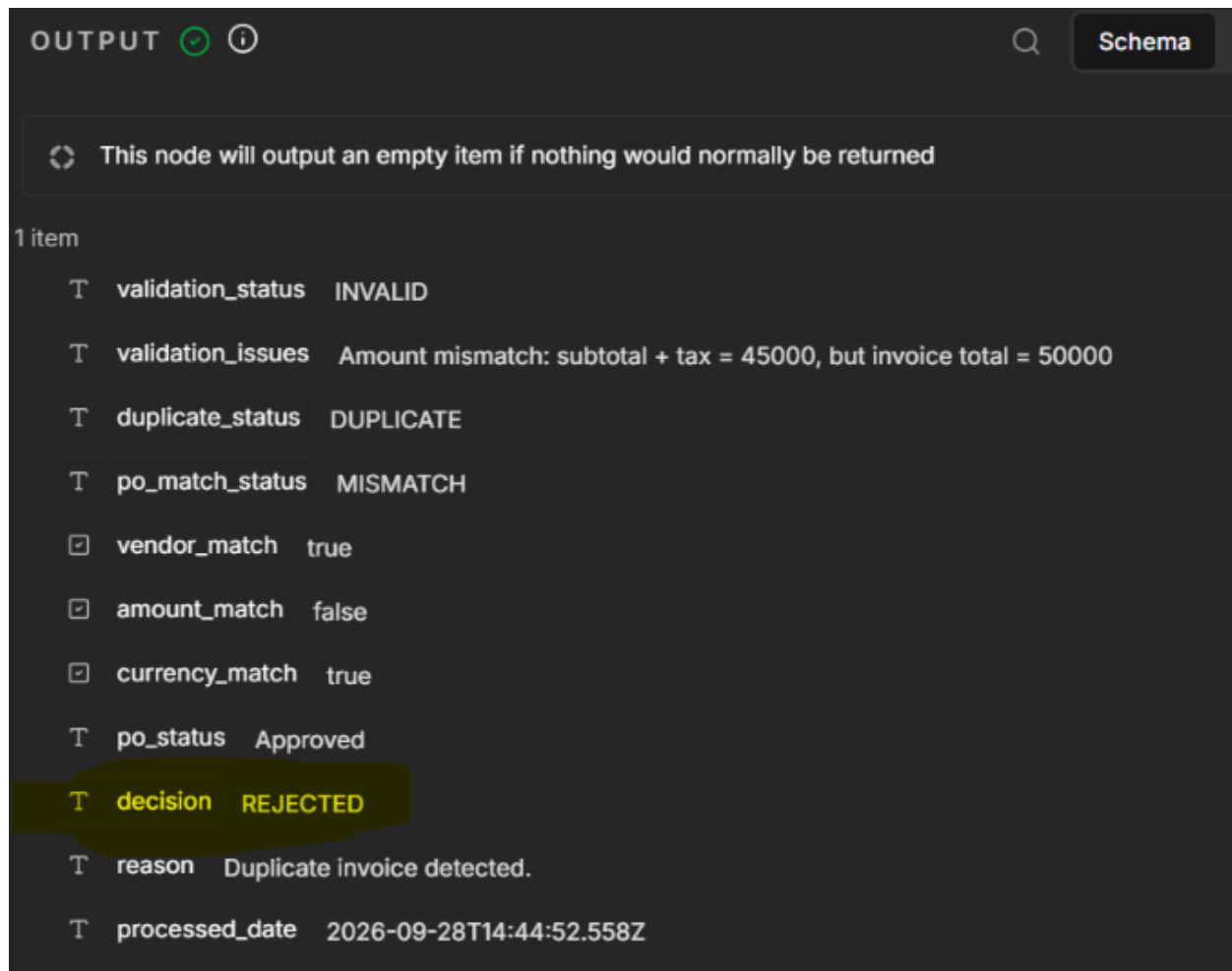


Figure 7. Decision Engine — Approved Case

13. Human-in-the-Loop

Invoices requiring review are not automatically approved. The routing node sends a Human Review Alert containing invoice details, validation status, duplicate status, PO match status, decision, reason, and validation issues.

SCREENSHOT 9 — Human Review Gmail Alert

Invoice INV-1008 - HUMAN REVIEW Inbox x

Shubham Vishwakarma

to me ▼

Hello,

The invoice processing workflow has completed successfully.

Invoice Number: INV-1008

Vendor: ABC Supplies Pvt Ltd

Invoice Date: 2026-09-28

PO Number: PO-501

Subtotal: ₹40000

Tax: ₹5000

Total Amount: ₹50000

Currency: INR

Validation Status: INVALID

PO Match Status: Approved

Decision: HUMAN REVIEW

Reason:

Amount mismatch: subtotal + tax = 45000, but invoice total = 50000

Processed Date:

2026-09-28T14:18:21.356Z

This notification was generated automatically by the AI Invoice Processing Agent.

This email was sent automatically with n8n

<https://n8n.io>

Figure 9. Human Review Gmail Alert

14. Processing Notification

Completed invoices are sent through the normal processing notification branch with invoice details, validation result, PO match result, decision, reason, and processed date.

SCREENSHOT 10 — Processing Notification

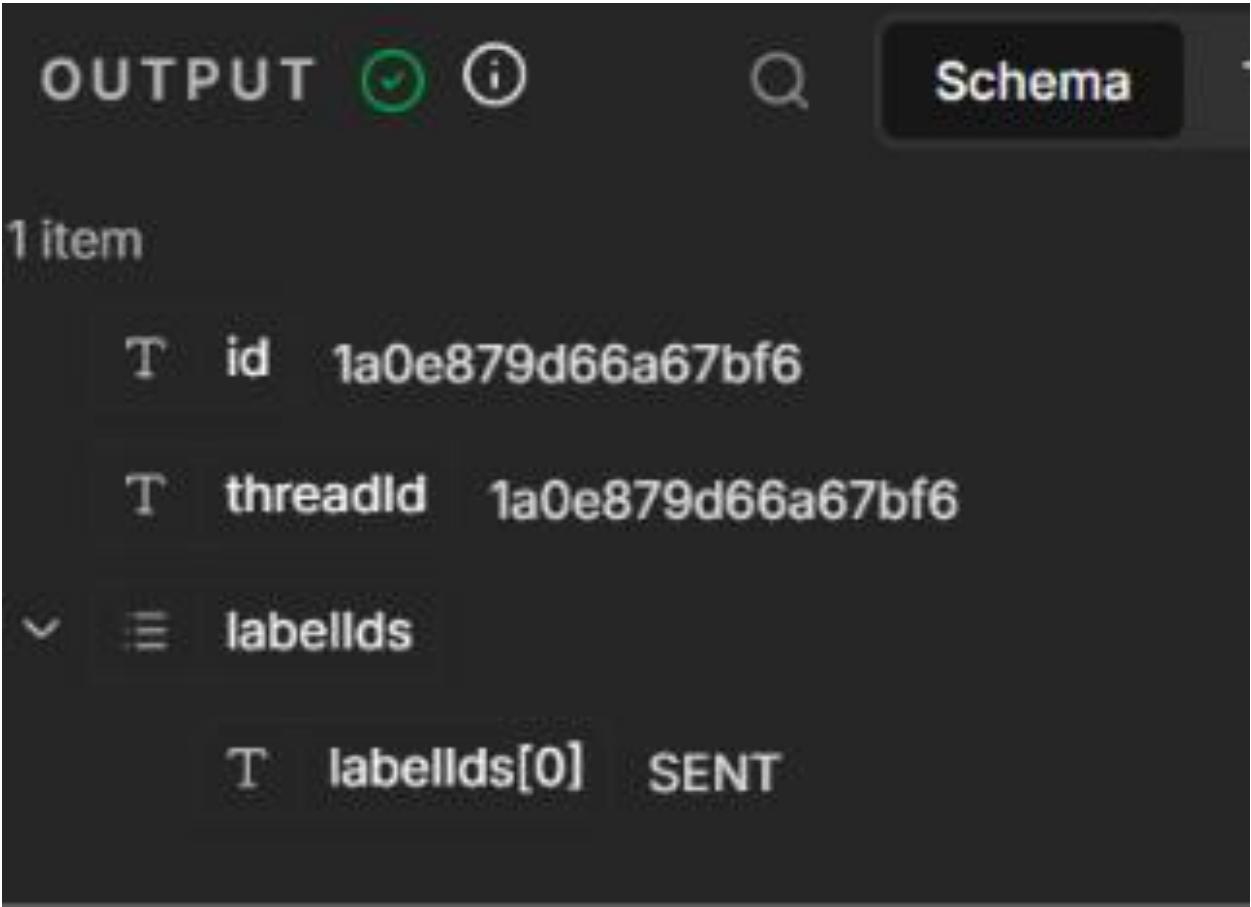


Figure 10. Processing Notification

15. Google Drive Storage

The original invoice PDF is uploaded to the designated Google Drive invoice folder, preserving the source document used by the workflow.

SCREENSHOT 11 — Original Invoice in Google Drive

My Drive > AI Invoice Processing - I... ▾

Ask Gemini | Type ▾ People ▾ Modified ▾ Source ▾

Name ▾	Owner	Date modified	File size	Sort
 Invoice_INV-1007.pdf	 me	19:47 me	27 KB	⋮
 Invoice_INV-1007.pdf	 me	20:04 me	27 KB	⋮
 Invoice_INV-1007.pdf	 me	20:14 me	27 KB	⋮
 Invoice_INV-1007.pdf	 me	20:08 me	27 KB	⋮
 Invoice_INV-1007.pdf	 me	20:12 me	27 KB	⋮
 invoice_INV-1006.pdf	 me	19:39 me	27 KB	⋮
 invoice_INV-1006.pdf	 me	19:38 me	27 KB	⋮

Figure 11. Original Invoice in Google Drive

16. Test Cases and Expected Results

Representative development tests included INV-1007 for a valid matching case, INV-1005 for an amount-mismatch/human-review case, repeated invoice numbers for duplicate detection, and INV-1008 for an arithmetic-mismatch scenario. Use the actual captured execution evidence for the final submission.

17. Final Evidence Checklist

- ☐ Final frozen n8n workflow canvas
- ☐ Gemini structured extraction output
- ☐ Invoice validation output
- ☐ Duplicate detection output
- ☐ PO/vendor validation output
- ☐ Approved decision output
- ☐ Human review decision output
- ☐ Human review Gmail alert
- ☐ Normal processing Gmail notification
- ☐ Google Sheets Invoice_Log evidence
- ☐ Google Drive original invoice evidence

18. Key Controls and Design Principles

LLM is used for information extraction, not uncontrolled financial approval.

Required fields are validated before downstream decisions.

Invoice arithmetic is checked deterministically.

Duplicate invoice numbers are checked against a persistent log.

PO vendor, amount, currency, and status are checked against reference data.

Exceptions are routed to human review.

Processing outcomes and reasons are logged for traceability.

Original invoice documents are stored in Google Drive.

19. Conclusion

The AI Invoice Processing Agent demonstrates how Agentic AI and workflow automation can be combined for a practical business process. It integrates document intake, AI extraction, deterministic validation, reference-data matching, duplicate detection, decision routing, persistent logging, document storage, and human review. The AI is focused on unstructured document understanding while deterministic controls govern financial processing decisions.

Test Case	Scenario	Expected Result	Evidence	Status
TC-01	Valid invoice + matching PO	APPROVED	Decision screenshot	Completed
TC-02	Invoice amount differs from PO	HUMAN REVIEW	Decision + alert	Completed
TC-03	Duplicate invoice number	REJECTED	Duplicate + decision	Completed
TC-04	Invoice arithmetic mismatch	HUMAN REVIEW	Validation + decision	Completed